Python Multi-Page Test Suite

PYMUPDF (FITZ) PARSING & LAYOUT BENCHMARK

DOC-VER: 2026.4-MULTI

1. EXECUTIVE SUMMARY & DOCUMENT SCOPE

This multi-page technical specification serves as an automated test target for PDF extraction libraries, with a primary focus on `PyMuPDF` (fitz). It spans multiple continuous pages to test pagination flow, multi-page table extraction, text bounding box calculation, vector drawing parsing, and link resolution across page boundaries [1].

When evaluating multi-page PDF processing scripts, key performance indices include page traversal efficiency, search string bounding box extraction, table header propagation, and glyph mapping for extended Unicode ranges [2].

2. EXTENDED SYSTEM CONFIGURATION

Test Suite Identifier: TS-MULTI-9082X

Target Engine: PyMuPDF v1.23+

Page Count Target: 3 Pages

Coordinate System: 72 DPI Points (PDF Top-Left)

Text Extraction Mode: Text / Blocks / Words / JSON

Table Extraction API: page.find_tables()

3. MULTI-ROW SYSTEM PERFORMANCE TABLE

The following tabular dataset contains structured telemetry logs to test table detection algorithms over extended page structures.

TASK ID	SUBSYSTEM MODULE	EXECUTION STATE	LATENCY (MS)	MEMORY (MB)	SCORE INDEX
TSK-101	Document Catalog Parser	COMPLETED	3.42	12.8	0.998
TSK-102	Page Flow Rasterizer	COMPLETED	18.90	45.2	0.985
TSK-103	Font Matrix Decoder	COMPLETED	6.15	18.4	0.999
TSK-104	BBox Search Indexer	COMPLETED	11.02	24.1	0.974
TSK-105	Table Boundary Detector	COMPLETED	22.40	52.6	0.961
TSK-106	Glyph Mapping Engine	COMPLETED	5.88	14.2	0.992
TSK-107	Annotation & Link Extractor	COMPLETED	4.10	9.6	1.000
TSK-108	Vector Drawing Stream Reader	WARNING	38.75	88.4	0.912
TSK-109	Content Stream Tokenizer	COMPLETED	7.30	16.0	0.995
TSK-110	Metadata Stream Resolver	COMPLETED	2.15	8.1	1.000
TSK-111	Multi-Column Layout Analyzer	COMPLETED	15.60	31.5	0.978
TSK-112	Unicode Normalize Transformer	COMPLETED	8.90	20.2	0.989
TSK-113	Pixmap Renderer (150 DPI)	COMPLETED	54.10	112.0	0.965

TASK ID	SUBSYSTEM MODULE	EXECUTION STATE	LATENCY (MS)	MEMORY (MB)	SCORE INDEX
TSK-114	Cross-Reference Table Inspector	COMPLETED	3.05	7.9	0.999

4. CHARACTER ENCODING & SCIENTIFIC NOTATION

Unicode Encoding & Mathematical Verification:

- **Currency Symbols:** $ (USD), € (EUR), £ (GBP), ¥ (JPY), ₹ (INR), ₿ (BTC)
- **Statistical Equations:** $s^2 = \sum (x_i - \bar{x})^2 / (n - 1)$
- **Matrix Representation:** $A = [a_{i,j}] \in \mathbb{R}^{m \times n}$, $\det(A) \neq 0$
- **Greek Symbols & Constants:** α = 0.05, β = 0.20, γ = 1.618, λ = 2.718, π ≈ 3.14159
- **Special Formatting Strings:** Left-to-right markers, curly quotes (“test”), non-breaking spaces, and em-dashes (—).

5. BOUNDING BOX & KEYWORD SEARCH TARGETS

To verify PyMuPDF's `page.search_for()` method, locate the exact coordinates of the following target phrase: "**CRITICAL_SEARCH_TARGET_2026**".

PyMuPDF should return a list of `fitz.Rect(x0, y0, x1, y1)` objects corresponding to the precise spatial location of the target string on this page.

6. BIBLIOGRAPHIC REFERENCES

1. PyMuPDF Engineering Team, "High-Performance Document Extraction Engine Technical Manual," *Journal of Open Source Software*, 2024.
2. McKenzie, L. & Zhao, R., "Benchmarking PDF Table Parsing Algorithms Across Complex Layouts," *Data Science Proceedings*, vol. 18, pp. 112–128, 2025.